MIF — Minimum Implementation Framework (AHIM™)

Module: AHIM™ — Agent Harness Integrity Module
Parent Standard: Runtime Integrity Standard (RIS)
Type: Minimum Implementation Framework

Purpose

This framework defines the minimum structural and operational conditions required to implement Agent Harness Integrity (AHIM™).

It ensures that AI agent systems remain controlled, verifiable, and valid during real-time execution.

Implementation Principle

Execution must remain controlled, bounded, and verifiable at all times.

Core Implementation Architecture

A system implementing AHIM™ must establish six non-bypassable control layers:

1. Execution Boundary Layer

The system must:

- define explicit execution boundaries
- prevent actions outside authorized scope
- enforce runtime constraints at all times

---

2. Tool Governance Layer

The system must:

- define permitted tools and actions
- enforce authorization for all tool interactions
- ensure all tool usage is traceable

---

3. Memory Integrity Layer

The system must:

- control read/write operations
- prevent unauthorized mutation
- maintain consistent and coherent state

---

4. Session Control Layer

The system must:

- define session lifecycle and boundaries
- control state transitions

- prevent uncontrolled persistence

5. Environment Control Layer

The system must:

- isolate execution environments
- prevent uncontrolled external effects
- ensure auditable system behavior

6. Outcome Verification Layer

The system must:

- link outputs to execution context
- enforce constraint alignment
- enable auditability of all results

Minimum Operational Requirements

To be considered AHIM™ compliant, a system must:

- enforce all control layers simultaneously
- maintain real-time visibility into execution
- detect and prevent deviation from defined conditions
- ensure that all actions are attributable and traceable

Implementation Steps

1. Define execution scope and boundaries
2. Establish tool authorization and governance
3. Implement memory control mechanisms
4. Define session lifecycle and limits
5. Isolate and control execution environment
6. Enable output traceability and verification

Continuous Enforcement Condition

All implementation layers must operate continuously during runtime.

If any layer becomes inactive or bypassed, system integrity is invalid.

Failure Condition

Implementation is invalid if:

- execution boundaries are not enforced
- tool usage is uncontrolled
- memory state is inconsistent
- sessions are unbounded

- environment allows uncontrolled impact
- outputs cannot be verified

System Output

A valid AHIM™ implementation produces:

- controlled execution behavior
- traceable and auditable actions
- consistent runtime state
- verifiable outputs

Canonical Closing Statement

A system is not defined by how it is built, but by how it behaves during execution.
If execution is not controlled, integrity does not exist.

Related Documents

→ AHIM™ — Agent Harness Integrity Module

→ AHIM™ — Self-Declaration Framework

→ Runtime Integrity Standard (RIS)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

https://originopenfoundation.org/